

PREPARING ACCOUNTING GRADUATES FOR DIGITAL REVOLUTION: A CRITICAL REVIEW OF INFORMATION TECHNOLOGY COMPETENCIES AND SKILLS DEVELOPMENT

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January 2016

Acknowledgments: This research was supported by the Singapore Ministry of Education (MOE) Academic Research Fund (AcRF) Tier 1 grant (13-C206-SMU-002).

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ABSTRACT

The pervasiveness of information technology (IT) in businesses has altered the nature and economies of accounting activities. In particular, the emergence of cloud computing, eXtensible Business Reporting Language (XBRL), and business analytics in recent years have transformed the way companies report financial performance and make business decisions. As a consequence, there is a surge in demand for advanced IT skills among accounting professionals. In this research, we aim to review selected articles published between 2004 and 2014 based on selected search phrases. The outcome of the review may serve as an important input for current and future accounting curriculum revisions.

Keywords: accounting graduates, information technology competencies, skills development, accounting information systems

INTRODUCTION

The accounting industry has experienced several turbulent cycles over the last decade. In particular, investors' confidence in the capital markets was severely dented when the avalanche of accounting scandals became public in 2002 (Adler, 2002). In order to revive investors' confidence, the Sarbanes Oxley Act (SOX) (SEC, 2002) was enacted in the United States to strengthen corporations' internal control systems. The enactment of SOX had far-reaching effects as it triggered the adoption of a more risk-based business approach, which called for a critical re-examination of business processes, internal control systems and supporting information systems (IS) infrastructure. Specifically, Sections 404 and 802 of SOX pose direct impacts on information technology (IT) investment. Section 404 aims at improving corporate governance and reporting with heavy emphasis on internal control systems within corporations. Section 802 requires all business records to be stored at least 5 years which may result in a surge of IT storage budgets. These governance regulations have triggered corporations to re-evaluate their current financial reporting systems and integrate their IT solutions within their reporting processes. As a result, SOX requirements have led to a higher level of corporate investment in IT.

Besides regulatory compliance, higher IT adoption rate among accounting functions is attributed to the push for improved productivity among accounting professionals. There has been increased investment in audit software and knowledge sharing applications to support accounting firms to gather, organize, process, evaluate and present financial data (Curtis et al., 2009). In addition, since teamwork is critical in a public accounting firm as audit engagements are performed by teams composed of professionals at different ranks, these accounting firms also invest in

groupware technology to improve work collaboration and communication within teams, thus enhancing accounting productivity (Vandenbosch and Ginzberg, 1997). Besides, the emergence of cloud computing, eXtensible Business Reporting Language (XBRL), and business analytics have transformed the way corporations report financial performance and make business decisions (Pan et al., 2015).

Due to increasing IT investment among accounting functions, it is therefore not difficult to understand why there is growing demand for advanced IT knowledge and skills for accounting professionals. According to some anecdotal evidence, advanced IT knowledge and skills such as IT forensic, IT audit and data analytics are clearly lacking in numbers among accounting professionals. The accounting industry urgently needs a group of sophisticated accounting information systems (AIS) professionals that could handle sophisticated IT demands in accounting services and activities. This calls for the need for accounting educators to review existing AIS curriculum and assess whether current curriculum is adequate to equip accounting students with advanced IT knowledge and skills (Arens and Elder, 2006; Arnold and Sutton, 2007; Curtis et al., 2009).

In order to develop a holistic AIS curriculum in response to the growing need for advanced IT knowledge and skills, we reviewed selected articles published in accounting journals and industry reports between 2004 and 2014 based on selected search phrases relating to IT skills and competencies necessary for accounting professionals. We performed content analysis on IT competencies and skills development in the articles and classified them into a number of knowledge areas that encapsulate the literature in the selected articles. The outcome of the review may serve as important input for current and future AIS curriculum revisions.

The review is timely especially when critics have clearly highlighted the inadequacies within current AIS curriculum in preparing accounting students for advanced IT knowledge and skills (AAA, 2012; AACSB, 2014; AICPA; Curtis et al. 2009; Evan et al. 2012). Curtis et al. (2009, p. 92) commented that a “renewed investment in education research on the IS preparedness of accounting students is necessary”. Evan et al. (2012) states that there are global concerns that the accounting curriculum fails to keep up with advances in technologies. Several accounting professional bodies have also commented on the importance of developing IS skills among accounting graduates. For example, the Pathways Commission Report released by the American Accounting Association (2012) indicates that one of the action items is to transform learning experiences to reflect current and emerging technologies in business. AICPA developed a Core Competency Framework, which defines a set of skills-based competencies needed by all students entering the accounting profession. Under the Core Competency Framework, AICPA states that accounting professional needs to demonstrate technological adaptability by acquiring new IT skills and determining how new technologies should be best incorporated into their accounting practices. Recent AACSB report (2014) has also commented on the need to include business analytics in the AIS curriculum in preparation for future accounting work. AACSB (2014) states that accounting degree programs need to integrate current and emerging accounting and business information technologies throughout the academic curricula. Overall, our literature review aims to address three research questions: (1) what are the impacts of IT on accounting, (2) what IS-related courses should be offered to

accounting students so as to equip them with the necessary IT competencies and skills? And (3) how to make these courses feasible?

The paper is organized as follows: In the next section, we will explain the research method. This is followed by the discussion of the impacts of IT on Accounting, a number of AIS knowledge areas and a list of proposed AIS courses. The remaining sections focused on issues to be considered and concluding remarks.

RESEARCH METHOD

The process of identifying and reviewing the literature involved two main steps: (1) identifying relevant articles to be examined and (2) determining the classification categories. Using the Business Source Complete Database, we conducted a search for publications whose titles, abstracts or keywords contained the selected search phrases by browsing major accounting, management and information systems journals. In terms of period selection, we focused on the period between 2004 and 2014. The period selection is due to a sharp increase of IT applications used in the accounting industry in the last decade, particularly in the areas of transaction processing, controls, audit and analytics.

The search phrases included ‘accounting information systems’, ‘IT for accounting professionals’, ‘computer audit’, ‘system control’, ‘digital forensic’, ‘XBRL’, ‘business analytics’ and ‘IT in accounting curriculum’. Abstracts of all articles identified were read to decide if they fall within the scope of the study. Altogether, 34 articles were reviewed. Refer to appendix A for the reference details of some of these articles. Some of the journals that were examined include: The Accounting Review, Critical Perspectives on Accounting, Journal of Accounting

Research, Accounting, Organizations and Society, Accounting Horizons, Accounting, Auditing and Accountability Journal, Managerial Auditing Journal, Journal of Forensic Accounting, Journal of Accountancy, Accounting Education, Issues in Accounting Education, Advances in Accounting Education, Journal of Information Systems and International Journal of Accounting Information Systems. Besides academic journals, we also examined major periodicals and relevant information from websites.

The data analysis involved the following steps: (1) define the phenomenon in a tentative manner using topics identified from the literature review, (2) scan the text to identify IT competency and skills development topics, and (3) refine competency areas until all articles are accounted for. Relevant texts from these articles were sorted according to IT competency and skills development areas. A list was compiled to contain the location of each relevant text on an article, the article reference, and links to other texts and articles. In order to reduce researcher bias, another colleague was asked to participate in early analysis of some of the data. The role of this colleague is to bring a different perspective in the data analysis and classification process.

Coding categories reflect our interpretations of the IT competency and skillsets. We conducted an initial pilot run for coder training and pilot testing of reliability. During the pilot run, we also refined the coding instrument and procedures. To establish the reliability of the coding, each coder was asked to quote a particular segment of the relevant texts. Coding was conducted independently and without consultation and guidance. We examined the portions of the coding where both coders agreed and measured the inter-coder reliability using Cohen's Kappa

coefficient. Our coefficient score of 0.77 suggests substantial agreement between the two coders, and the result also demonstrates that the categories were clearly defined and could be located in the text with little ambiguity. As the reliability coefficient was high, each coder was subsequently asked to code separate portions of the texts.

DISCUSSION

Impacts of IT on Accounting

Undoubtedly, accounting, being a discipline concerning disclosure of, or provision of assurance on accurate financial information, was influenced by accounting functions' growing dependence on technologies in business transactions (Gelinas and Gogan, 2006). Continuing developments in IT are likely to affect various aspects of accounting (Lord, 2004). For example, firms lever off the impacts of IT by modifying their business models. They can further automate their business processes and incorporate technologies, such as web services architecture and Internet-based supply chain management systems, into their infrastructure. The nature of these technology-oriented "assets" is different from physical assets, such as factory and equipment. Accountants are usually able to value physical assets reliably and their values do not change significantly within an accounting period. The evidence of the rapid change in the capital market of the values for firms in an information-driven market (e.g. Dell) indicates that their underlying technology-oriented assets change significantly (Li and Alam, 2007). Under AASB 136 Impairment of Assets, accountants and auditors have a legal obligation to reliably record a non-current asset to its recoverable amount when its carrying amount is greater than its recoverable amount. Hence, accountants and auditors must have

knowledge of technology infrastructures and their impacts on business processes (Davern et al., 2005). There are four knowledge areas that include Internal Control, IT Control and Auditing, Data Modelling, Tagging and Management, and Data Analytics.

Internal Control

For business processes, accounting professionals have to learn how to examine and interpret the embedded rules in enterprise resource planning systems that influence cost allocations. Given the requirements of SOX Act of 2002, internal audit functions must use appropriate technology to increase their efficiency and effectiveness. Internal controls are enacted within enterprise resource planning systems to execute the calculations and record transaction entries to ensure data validity, accuracy and completeness. Technical skills and knowledge become a must for auditors to understand and implement the system control process. As a result, professional accounting bodies, such as the International Federation of Accountants (IFAC), strongly support the integration of the study of IT-related courses with accounting programs.

IT Control and Auditing

With the adoption of enterprise resource planning systems, firms transfer and store data in electronic form, and most transactions are performed online (Rikhardsson and Kræmmergaard, 2006). Accountants are expected to know not only how to use basic accounting software to manage data, but also have a solid understanding of critical technological components that drive information systems.

Technology-related risks and security issues are drawing auditors' attention. Identity management is now the primary transaction security focus for banks and financial institutions. IT controls and audits have to be implemented properly for diagnosing vulnerabilities and threats to firms (Weidenmier and Ramamoorti, 2006). In addition, accountants and auditors play a significant role in information technology governance, which is to ensure that business systems deliver the values that the organization board expects (Ferguson et al., 2013). Thus, accountants and auditors are expected to have a very high level of technology-related knowledge and grounding.

Data Modeling, Tagging and Management

In recent years, eXtensible Business Reporting Language (XBRL) has changed the financial reporting process (Farewell, 2006). By tagging individual pieces of business information with common data definitions and providing a means of transporting the information over the Internet, XBRL increases the speed and the consistency of financial data and the data capturing process. Traditionally, accountants have been in the vanguard of turning a largely untested financial reporting framework into a basis for financial reporting in more than 90 countries. Many countries in Europe and Asia are adopting and using International Accounting Standards (IAS). The way in which XBRL describes financial and business reporting data helps firms to enhance the transition to a full alignment for those countries currently using their local standards to transition to IAS. With the changing regulatory requirements affected by, for example, the Sarbanes-Oxley Act, legislation in the US International Financial Reporting Standards (IFRS), the role and

tasks of accountants and auditors are changing significantly. As a result, XBRL is seen not only a technology advancement, but also a change of regulations and practices in accounting and auditing.

XBRL enables the disclosure of insider trading information, which is currently available in electronic form on Electronic Data Gathering and Retrieval (EDGAR) through the Securities and Exchange Commission (SEC) website (Debreceeny et al., 2005). Historically insider trading transactions were publicly disclosed by an SEC Form 4 filing 10 days after the end of the month of the trade. As required by Section 403 of the Sarbanes Oxley Act of 2002, insider trading information is now disclosed within 48 hours of the insider trade in the XBRL format. New reporting analysis frameworks for general users to retrieve and analyze XBRL files from the SEC should be proposed by accountants with reasonably good technology-related knowledge. Since software tools that they use to prepare statements are able to import and publish XBRL-tagged documents much in the same way as HTML, accountants are not expected to master the technical infrastructure of XBRL standard. However, they do need to understand the XBRL filing process in order to learn its impact on accounting and audit procedures.

Data Analytics

Furthermore, as timeliness of information is key to the work processes in the financial sector, accountants will have to learn how sophisticated information systems such as business analytics and data mining, may help them in their work. Essentially, technology advances have enabled auditors to work more efficiently and made some aspects of the audit work less tedious. The use of business analytics has

allowed companies to plough through large amounts of data and identify outliers more accurately and quickly, compared to the old days when auditors had to sample items and do things manually. The digital age brings with it new challenges and opportunities for the accountant. It has transformed the way auditors think from looking backward to looking forward-calculating and forecasting the future. Going forward, accountants are no longer focused on resolving issues on back-end accounting processes. There is an opportunity for accountants to propose solutions that are more strategic-looking at end-to-end process improvements that cut across various units and departments. Process improvements are no longer focused on processes, but are instead geared towards improving the customer's experience so that they are aligned with global best practices. There is also greater transparency now with the help of technology. Key stakeholders are given close to real-time data to make timely decisions to support the growth and pace of business. Tools are also used to improve forecast accuracy, so that the right decision can be made in advance. Accounting professionals have the opportunity to leverage on the new tools in the digital age to respond faster and at lower costs, hence allowing resources to be distributed to other areas that yield significant topline returns to the business.

Proposed AIS Model Curricula

In this study, we have developed an Accounting Information Systems model curricula for the undergraduate level. The model curricula are based on the needs and expectations of information systems skills of the accounting profession which enable accountants to develop knowledge to manage IT and operate effectively in the changing technological environment. The model consists of knowledge areas that

include Internal Control, IT Control and Auditing, Data Modelling, Tagging and Management, and Data Analytics. Refer to Figure 1 below.

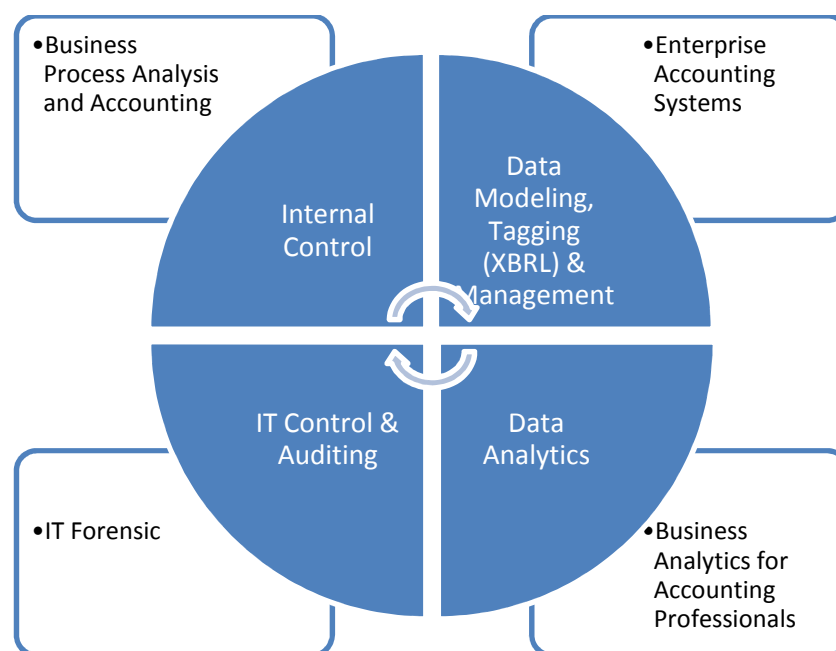


Figure 1: Accounting Information Systems Model Curricula

Together, these knowledge offer a set of skills-based competencies needed by all students entering the accounting profession, regardless of the career path they choose or the specific accounting services they will perform.

Proposed AIS Courses

Today's accounting students are expected to learn some technology-related concepts (Fordham, 2005). Disappointingly, some accounting students misunderstand that technologies refer to the basic use of accounting information systems, electronic commerce, or even spreadsheet and word processing software. They think that they can learn fundamental and in-depth IT knowledge after

graduation. Obviously, a clearly-presented curriculum structure and relatedness between accounting and IS is the first step to influence accounting students' aspirations to pursue AIS courses. IS courses should be customized to the accounting background (Doost et al., 2011). To tailor to the needs of accounting students, the courses should be business and non-technically oriented. This can not only be a better fit with students' accounting background, but also strengthen students' confidence in their ability to acquire relevant knowledge and perform well in these courses.

We sampled 75 university websites from the ISWorld list. Amongst these universities, about 15 of them merged accounting and IS into one department and offered an accounting programme which consisted at least one entry-level AIS course. To a certain extent, some programmes provided accounting students with an elective course, project management, probably the rationale behind was that the new role of accountants would be "information managers". A plausible approach to impart relevant technical skills to accounting students is to embed IS knowledge within an accounting context. By demonstrating how IT plays a significant role within an accounting setting, one may solicit greater buy-in from accounting students.

We propose to provide accounting students with a fully-fledged accounting information systems curriculum at the elementary, intermediate and advanced levels. At the elementary level, a course may focus on providing an integrated view on information systems and related accounting processes in organizations. It can provide students with a broad knowledge of the fundamentals of accounting data management, information integration, and modeling of business processes. At the

intermediate level, a course may be offered to examine how organizations implement enterprise systems to link various business functional systems into a tightly integrated single system with shared data and how accountants may utilize the enterprise resource planning systems to organize and present updated financial information. After the students gain a general idea of auditing in their second year or third year, at the advanced level, a course may be designed for accounting students to better understand computer frauds and fraud detection methods. Given the statement on Auditing Standards (SAS) No. 99 (Consideration of Fraud in a Financial Statement Audit) (AICPA), auditors are required to become more vigilant in assessing fraud risk. Thus, this course is closely related to auditing. Another advanced level course may be 'business analytics for accounting professionals'. With complexity and data proliferation, increasingly the CEO and the board turn to accountants to help make sense of all the data. Therefore, business analytics may help cut through this complexity, and to provide more informed analysis on the business and its operation.

In the following sections, we specify four courses: *Business Process Analysis*, *Enterprise Accounting Systems*, *Information Technology Forensics* and *Business Analytics for Accounting Professionals* for elementary, intermediate, and advanced levels respectively. We also provide a short synopsis of what these courses entail.

Business Process Analysis and Accounting

This course equips accounting students with basic database and documentation skills, and helps them gain an understanding of accounting processes, including ordering, purchasing, production and human resources, and data stores.

Basic database skills include relational database, referential integrity and form design to eliminate user input errors. Structured query languages can further enhance their knowledge on extracting and querying the data. Regarding basic documentation skills, accounting students should learn how to document a narrative of accounting processes with techniques, such as data flow diagrams and system flowcharts. Analytical skills of assessment and evaluation of the risks associated with accounting information systems are highly relevant to auditing. Students should learn how to design and evaluate control practices to manage those risks, to improve the internal control and to achieve information validity, accuracy and completeness. Greater weight should be placed on internal control and transactions processing, whereas moderate importance should be placed on software and hardware issues (Doost et al., 2011).

Information Technology Forensic

Apart from corporate fraud and disputes, organizations face the risks of their information systems being compromised. Risks include computer frauds, virus attacks and server hackings. As such, organizations ought to take every precaution to protect their systems through tighter internal control and audits. Accounting students who aspire to be future auditors or forensic accountants may need to understand how computer frauds may be conducted, how to increase the difficulty of committing a fraud, and ways to improve fraud detection methods. Accounting students should be equipped with basic technology-related knowledge, such as network technologies and the Internet infrastructure, to know the IT-related risks, their perpetration, investigation, and prosecution. This course puts an emphasis on the legal framework

and the investigation of fraud, principles of handling computer evidence, analysis of computer evidence, and presentation of evidence in a court as an expert witness. As data analytics becomes more prevalent, the use of advanced analytical technologies to identify fraud may increase. Traditionally, corporate fraud detection tends to rely heavily on the manual skills of the fraud investigation team using experience, instinct, and persistence to analyze data related to the fraudulent activity. Nevertheless, the sheer volume, velocity and variety of data that are now being generated every day in a corporation, are likely to overwhelm fraud investigators' best attempts at data analysis. Therefore fraud detection has to evolve to be more technology-centric so as to make better sense of big data insights.

Enterprise Accounting Systems

Enterprise resource planning systems are considered to be one of the most important technologies in accounting or auditing field (Strong et al., 2006; Sutton, 2004). Relative to the earlier course, this course is more technology-oriented. Students should gain an understanding of the impact of various e-business technologies. For instance, Greenstein and McKee (2004) carried out a literature search and came up with 36 important information technologies, with the confirmation from more than 300 AIS professors and 200 auditing practitioners. Apart from basic computer skills (e.g. database searching, word processing and spreadsheets), sophisticated areas are also expected. Sophisticated areas include expert systems, encryption software, groupware, computer security and intrusion detection and enterprise resource planning systems. Standard practices to establish control over development, implementation and maintenance of accounting

information systems should be covered. This course should aim to enhance students' knowledge in advanced technologies, including XBRL, data mining and data warehousing. Data mining and data warehousing are relevant to accounting disciplines, because auditors use technologies to deal with large amounts of financial data for fraud detection.

Business Analytics for Accounting Professionals

The course focuses on how an organization can leverage information through business analytics to achieve its goals. Business analytics is the exploration and investigation of past business performance to gain insight to existing performance and to assist in future planning. There is growing evidence to suggest that investment in Business Analytics can improve company competitiveness. In this course, students learn how to identify the information needs of an organization as well as information that would drive competitive advantage. Students would also explore the tool box of technologies and best practices available and learn how to best harness them. With complexity and data proliferation, increasingly the CEO and the board turn to accountants to help make sense of all the data, to help cut through this complexity, and to provide more informed analysis on the business and its operation. The opportunity for accounting function is if it can generate the insights that help make better corporate decision making, while continuing to ensure effective control of the enterprise, its reputation as a catalyst for growth will be guaranteed.

Issues to be considered

As AIS courses are on the boundary across two disciplines, accounting and IS, a critical issue is whether these AIS courses should be taught by accounting professionals or IS professionals? Should this content be taught in a standalone course or embedded in some existing accounting courses? To teach effectively, instructors should have reasonable accounting knowledge to contextualize the IT content in accounting and audit domains. A tight integration of IS courses into the accounting curriculum is required. Our suggestion is different from a typical approach adopted by some commerce or business programs, in which an introductory IS course is offered to all first year commerce students. Nevertheless we do acknowledge customizing courses for accounting students may incur more teaching preparation effort.

In addition, the support by professional bodies and accountancy organizations is crucial to leverage our efforts. Accounting programs in tertiary education recognize the importance of IT, and some have tried to enrich the AIS curriculum by exposing students to developments in IT. For instance, in 2001, the education committee of IFAC asserted that “information technology is pervasive in the world of business. Competence with this technology is an imperative for the professional accountant.” In January 2003, IFAC published an article to highlight the importance of information technology for professional accountants. To strongly promote AIS courses to accounting students, we need support from the accounting and auditing communities. For instance, professionally accredited programs, such as Certified Practicing Accountant (CPA) Program, currently include modules that assume candidates have a mastery of not only accounting, auditing and taxation law, but also

finance. These modules primarily focus on advanced 'accounting and finance' issues, leaving little room in the program for modules that advance IT knowledge. Accounting students need to complete these courses in order to gain their CPA qualification. To recognize the significance of technology, these professionally certified programs should introduce more IT-related materials, and include them as part of certification completion criteria.

Finally, greater effort must be made by accounting industry to highlight the importance of IT in accounting-related jobs. For example, with the emergence and rapid expansion of forensic accounting and business analytics, there is a great need for data mining skills, as well as forensic IT investigative skills. Also, public accounting firms should try to fill positions with students having some exposure to SAP R/3. Consequently, the market needs to clearly signal the need for these skills in any job advertisement and position description.

CONCLUSION

IT is heavily used in accounting functions. Technologies, such as forensic tools, XBRL, analytics and data mining, are commonly used in accounting and audit sectors. Therefore, it is important for accounting students, the future accountants and auditors, to receive proper technologies training in their tertiary education, particularly AIS curriculum. Academics, practitioners and professional societies should help accounting students fully understand the importance and usefulness of AIS courses in their long term career. This paper also proposes four AIS courses at the undergraduate level, and points out several important issues for educators before offering these courses.

REFERENCES

- Adler, P (2002). "Corporate Scandals: It's Time for Reflections in Business Schools," *Academy of Management Executive*, (16)3, pp. 1-12.
- AAA, American Accounting Association, Pathways Commission, "Charting a National Strategy for the Next Generation of Accountants", Sarasota, Florida, July, 2012.
- AACSB, Association to Advance Collegiate Schools of Business, "AACSB International Accounting Accreditation Standard A7: Information Technology Skills and Knowledge for Accounting Graduates: An Interpretation", AACSB International Accounting Accreditation Committee, September, 2014.
- American Institute of CPAs, AICPA, "Core Competency Framework & Educational Competency Assessment" Retrieved on 4th October 2015 at <http://www.aicpa.org/InterestAreas/AccountingEducation/Resources/Pages/CoreCompetency.aspx>.
- American Institute of CPAs, AICPA, "Statement on Auditing Standard No. 99 Consideration of Fraud in a Financial Statement Audit" Retrieved on 12th February 2016 at <http://www.aicpa.org/Research/Standards/AuditAttest/DownloadableDocuments/AU-00316.pdf>
- Arens, A.A., and R.J. Elder (2006), "Perspectives on auditing education after Sarbanes-Oxley", *Issues in Accounting Education* (21)4, pp. 345-362.
- Arnold, V. and S.G. Sutton (2007). "The impact of enterprise systems on business and audit practice and implications for university accounting education", *International Journal of Enterprise Information Systems* (3)4, pp. 1-21
- Curtis, M.B., J.G. Jenkins, J.C. Bedard and D. R. Deis (2009), "Auditors' Training and Proficiency in Information Systems: A Research Synthesis", *Journal of Information System* (23)1, pp. 79-96
- Davern, M., C. Ferguson, and M. Pinnuck, (2005), "The Pervasiveness of Information and Communication Technology: Its Effects on Business Models and Implications for the Accounting Profession," *Australian Accounting Review* (15)3, pp. 39-43.
- Debreceeny, R.S. et al. (2005), "Financial Reporting in XBRL on the SEC's EDGAR System: A Critique and Evaluation," *Journal of Information Systems* (19)2, pp. 191-210.
- Doost, R., McCombs, G., and Sharifi, M (2011) "The State of Teaching Accounting Information Systems: Is There a Gap?" *Review of Business information Systems* (7)3, pp. 61-70.
- Evans, E., R. Burritt and J. Guthrie, "Emerging Pathways for the Next Generation of Accountants, The Institute of Chartered Accountants in Australia and Centre for Accounting, Governance and Sustainability, University of South Australia, Australia, June 2012.
- Farewell, S.M. (2006), "An Introduction to XBRL through the Use of Research and Technical Assignments," *Journal of Information Systems* (20)1, pp. 161-185.

- Ferguson, C., Green, P., Vaswani, R., and Wu, Gang. (2013), "Determinants of Effective IT Governance," *International Journal of Auditing* (17)1, pp. 75-99.
- Fordham, D.R. (2005), "New Roles for AIS Courses: A Surprising Finding from a Case Study," *Journal of Information Systems* (19)1, pp. 113-129.
- Gelinas Jr., U.J., and J.L. Gogan (2006), "Accountants and Emerging Technologies: A Case Study at the United States Department of the Treasury Bureau of Engraving and Printing," *Journal of Information Systems* (20)2, pp. 93-116.
- Greenstein, M., and T.E. McKee (2004) "Assurance Practitioners' and Educators' Self-Perceived IT Knowledge Level: An Empirical Assessment," *International Journal of Accounting Information Systems* (5)2, pp. 213-243.
- Li, W., and P. Alam (2007), "Information Technology Capability: Firm Valuation, Earnings Uncertainty, and Forecast Accuracy," *Journal of Information Systems* (21)2, pp. 27-48.
- Lord, A.T. (2004) "ISACA Model Curricula," *International Journal of Accounting Information Systems* (5)2, July 2004, pp. 251-265.
- Pan, G., P.-S. Seow, C. Chan and C.-Y. Lim (2015), "Analytics and cybersecurity: The shape of things to come", CPA Australia, Singapore
- Rikhardsson, P. and P. Kræmmergaard (2006), "Identifying the Impacts of Enterprise System Implementation and Use: Examples from Denmark," *International Journal of Accounting Information Systems* (7)1, pp. 36-49.
- Strong, D., J. Fedorowicz, J. Sager, G. Stewart and E. Watson. (2006), "Teaching with Enterprise Systems," *Communications of the Association for Information Systems* (17), pp. 728-755.
- Sutton, S.G. (2004), "Rediscovering Our IS Roots," *International Journal of Accounting Information Systems* (5)1, May 2004, pp. 1-4.
- U.S. Securities and Exchange Commission, SEC (2002), "Sarbanes-Oxley Act of 2002" Retrieved on 12th February 2016 at <https://www.sec.gov/about/laws/soa2002.pdf>
- Vandenbosch, B. and M. J. Ginzberg. (1997), "Lotus Notes and Collaboration: Plus ça change," *Journal of Management Information Systems* (13)3, pp. 65-82.
- Weidenmier, M.L., and S. Ramamoorti (2006), "Research Opportunities in Information Technology and Internal Auditing," *Journal of Information Systems* (20)1, pp. 205-219.

Appendix A: List of Articles Relevant to IT Competencies and Skills for Accounting Professionals

Article	Journal	Author	Major Findings/Contributions	Competency (IT and skills)
An Examination of Audit Information Technology Use and Perceived Importance	Accounting Horizons. Mar 2008; 22(1); pp. 1-21	1. Diane Janvrin 2. James Bierstaker 3. Dr. Jordan Lowe	Results indicate that auditors extensively use a variety of audit applications including analytical procedures, audit report writing, electronic work papers, Internet search tools, and sampling. IT specialists use is infrequent, even for clients with complex IT. Findings also suggest that audit IT use and perceived importance vary by firm size.	Audit Applications such as audit planning software, digital analysis and fraud review. IT Productivity Tools: email, personal digital assistants, cell phones.
An Examination of Topical Coverage for the First Accounting Information System Course	Journal of Information Systems. Fall 2002, 16(2); pp 143-164	1. Craig E. Bain 2. Alan I. Blankley 3. L. Murphy Smith	Both faculty and professionals agree that greater importance should be placed on teaching internal control and transactions processing, while moderate importance should be placed on software and hardware issues. Professionals also placed higher importance on teaching software applications (particularly spreadsheet applications) than did faculty.	Skills: Internal control and transaction processing Software: spreadsheets, database applications, G/L applications
An Information Systems Auditor's Profile	Alternation. 2009, 16(1) pp. 318 - 335	1. Mariana Carroll 2. Alta van der Merwe 3. Sam Lubbe	Audit knowledge needs to be applied to IT knowledge. The audit tools and techniques are regarded as the supporting functions available to assist the IS auditor in performing IS audits.	IT knowledge: Application programs / ERP systems. IS Audit Tools: Generalized audit software (e.g. ACL, IDEA, Microsoft Excel and in-house developed SQL queries and scripts)
Assurance practitioners' and educators' self-perceived IT knowledge level: an empirical assessment	International Journal of Accounting Information Systems. 2004, 5, pp. 213– 243	1. Marilyn Greenstein 2. Thomas E. McKee	A relatively low level of knowledge for e-commerce and advanced technologies and audit automation constructs by both educators and practitioners was found, but there is a relatively high level of knowledge for office automation and accounting firm office automation constructs.	IT: Word processing, Electronic spreadsheets, E-mail, Electronic working papers, Internet search and retrieval, Image processing, Electronic presentations, , Embedded audit modules, Groupware, , Database design and installation
Computer-assisted audit tools and techniques: analysis and perspectives	Managerial Auditing Journal. 2003, 18(9), pp. 725-731	1. Robert L. Braun 2. Harold E. Davis	ACL is highly favored for improving both effectiveness and efficiency; however, documentation practices of using ACL need to be improved. Auditors expressed a desire to increase their skills through increased ACL training.	CAATS: Audit Command Language ACL and Interactive Data Extraction and Analysis IDEA

Article	Journal	Author	Major Findings/Contributions	Competency (IT and skills)
Critical information and communication technology (ICT) skills for professional accountants	Meditari Accountancy Research, 2005, 13, pp. 87-103	PL Wessels	This paper provides a list of critical information and communications technology skills required by accountants entering the profession.	Computer-assisted audit techniques, Database software, Internet tools, Professional research tools, accounting packages
E-commerce impact: emerging technology - electronic auditing	Managerial Auditing Journal, 2005, 20(4), pp. 408- 421	Junaid M. Shaikh	Enterprise architect can help create a common interface to integrate applications, allowing certain audit task to be conducted electronically over the internet	Embedded audit modules, System control audit review file (SCARF)
Employing generalized audit software in the financial services sector: Challenges and opportunities	Managerial Auditing Journal, 2005, 20(6), pp. 605- 618	1. Roger Debreceeny 2. Sook-Leng Lee 3. Willy Neo 4. Jocelyn Shuling Toh	The research finds that the extent and range of use of generalized audit software varies widely between the institutions.	ACL, IDEA and Panaudit Plus
Factors Affecting the Role of Internal Auditor in the Protection of Computerized Accounting Information Systems from Electronic Penetration	International Research Journal of Finance & Economics. 2011, 68, pp. 140-160.	Sharairi, Jamal Ade	There is an impact of knowledge and skill of the internal auditor on his role in the protection of computerized accounting information systems.	Internal auditors have to be adapt at electronic penetration (hacking) and testing the security controls against the contained information with an unknown source.
Identifying Employer Needs from Accounting Information Systems Programs	Journal of Information Systems Education. 2008, 19(4), pp. 403-410	Thomas W. Dillon	Companies look for a broad range of skills that integrate accounting and IT abilities and concepts into wide-ranging business knowledge.	Computerized AIS/EDP; Data modeling (EDP); ERP System (SAP, PeopleSoft); OS (Unix, Windows)
IT Instruction Methodology and Minimum Competency for Accountancy Students	Journal of Information Systems Education, 2002, 12(4) pp. 213	1. Jackson, Robert 2. J Owen Cherrington	Students felt that IT skills are necessary and important. They also indicate they would like more, not less IT education.	Spreadsheets, word processing, email, presentation Important: OS, Internet research, Database access, query, Accounting package
Accounting Students' IT Application Skills over a 10-year Period	Accounting Education: an international journal. 2009, 18(1), pp. 7-31	Gregory Stoner	IT skills constitute a significant element of the employability agenda. IT application skills with which students enter their university accounting studies are continuing to improve through time.	Database skills are likely to be one of the most important IT skills areas for the future of accounting.

Article	Journal	Author	Major Findings/Contributions	Competency (IT and skills)
Integrating Information Technology (IT) into Accounting Courses	International Journal of Business and Management. 2011, 6(10)	1. Ahmad N. 2. Obaidat, Rateb 3. M. Alqatamin	The results indicated that accounting students consider the four IT tools necessary for the learning process MS Excel earned the highest positive perceptions by accounting students followed by Internet, MS PowerPoint, and Email.	Spreadsheet, internet, presentation software and email
A Critical Learning Outcome Approach in Designing, Delivering and Assessing the IT Knowledge Syllabus	Accounting Education: an international journal. 2010, 19(5), pp. 439-456	Philippus Lodewikus Wessels	It is important for the educational process to encourage students to adopt a deep approach to learning, and that the focus should be on understanding rather than on memorization.	One of the requirements of IEPS 2 (IFAC, 2007, p. 6) is that accountants should obtain general knowledge of IT. The general IT knowledge forms the foundation for IT control knowledge and competencies.
The Information Systems-Accounting Nexus: Lessons from an Australian Institution	Communications of the Association for Information Systems. 2008, 22(11), pp. 197-210	1. Shuk Ying Ho 2. Gary Pan 3. Colin Ferguson	Results suggest that most accounting students expressed a weaker preference for selecting IS courses than finance courses.	Accountants are expected to know not only how to use basic accounting software to manage data, but also have a solid understanding of critical technological components that drive information systems.